

QI WANG

CONTACT INFORMATION

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EDUCATION

Ph.D. in Statistical Science, University of California, Santa Cruz

Expected May 2026

M.S. in Statistics, University of California, Irvine

Sept. 2019 - May 2021

B.S. in Applied Statistics, Central University of Finance and Economics

Sept. 2015 - July 2019

RESEARCH INTERESTS

Bayesian statistics, spatial and spatio-temporal processes, time series analysis, machine learning, and statistical deep learning for complex data

PUBLICATIONS

- **Spatial Deep Convolutional Neural Networks**
(Wang, Q., Parker, P.A. and Lund, R., 2025. Spatial deep Convolutional Neural Networks. *Spatial Statistics*, 66, p.100883.)
- **Hierarchical Count Echo State Networks With Application to Graduate Student Trends**
(Wang, Q., Parker, P.A. and Lund, R., 2025. Hierarchical Count Echo State Network Models with Application to Graduate Student Enrollments. *Data Science in Science*, 4(1), p.2565243.)
- **A Spatial-Temporal Study of Planetary Snow Trends**
(Under review at *Journal of the American Statistical Association*)

ONGOING RESEARCH PROJECTS

- **A Unified Copula Approach to Dependent Data Modeling**
 - Developed a deep learning approach for estimating marginal and joint cumulative distributions of dependent data (spatial, temporal, or spatio-temporal) using a Gaussian copula.
- **Change Point Detection for Cell Populations Measured by Flow Cytometry**
 - Utilized neural networks to detect change points in multivariate Gaussian mixture data.
- **Bridging Probability and Nonprobability Sampling: A Review of Methods, Theory, and Integration Strategies**
 - Reviewed theoretical frameworks and recent advances for making valid inferences using nonprobability and integrated survey data.

PEER REVIEW EXPERIENCE

- *Journal of Statistical Computation and Simulation* (1 Article)
- *Renewable Energies, Environment & Power Quality Journal (REE&PQJ)* (1 Article)

TEACHING

- **Graduate Student Instructor:** UCSC STAT/CSE-266A (Graduate Level Data Visualization and Statistical Programming in R), Spring 2025.
- **Graduate Student Instructor:** UCSC STAT-07L (Statistical Methods for the Biological, Environmental, and Health Sciences Laboratory), Summer 2021.
- **Teaching Assistant:** UCSC STAT-07 (Statistical Methods for the Biological, Environmental, and Health Sciences), Fall, Winter, Spring 2021, and Winter 2025.
- **Teaching Assistant:** UCSC STAT-131 (Introduction to Probability Theory), Fall 2023.

AWARDS

- UCSC Dissertation-Year Fellowship (\$33,000 plus tuition for the academic year), Spring 2025
- UCSC First Generation Graduate Group Research and Travel Award, Spring 2025
- UCSC Graduate Dean's Research Travel Grant, Winter 2025
- UCSC Regents Fellowships, Spring 2022

TALKS

- (Topic-contributed Session) 2025 Joint Statistical Meetings(JSM), Hierarchical Count Echo State Network Models with Application to Graduate Student Enrollments in “Hierarchical Modeling and Machine Learning for Complex Survey Data”
- International Symposium on Forecasting 2025, Hierarchical Count Echo State Network Models with Application to Graduate Student Enrollments.
- (Contributed Session) 2024 Joint Statistical Meetings(JSM), Spatial Deep Convolutional Neural Networks in “Recent Advances in Machine Learning Based Methods for Spatio-Temporal Analysis”

PROFESSIONAL AFFILIATIONS AND SERVICE

- Member, American Statistical Association (ASA)
- Member, Eastern North American Region (ENAR)